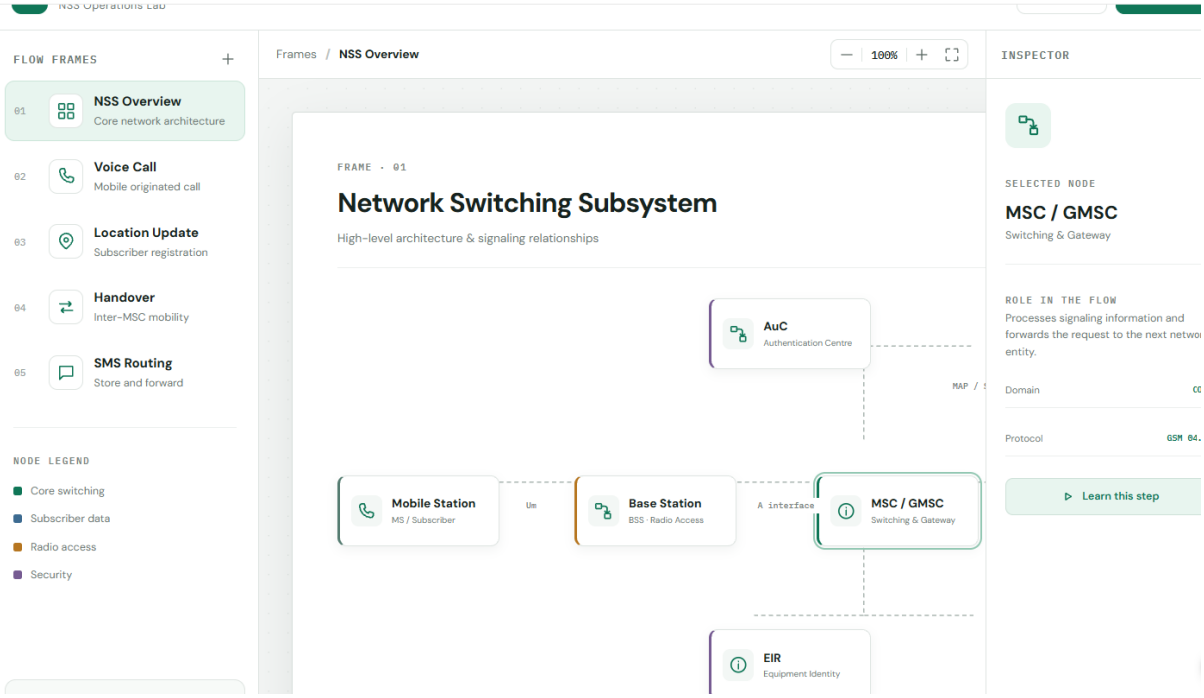
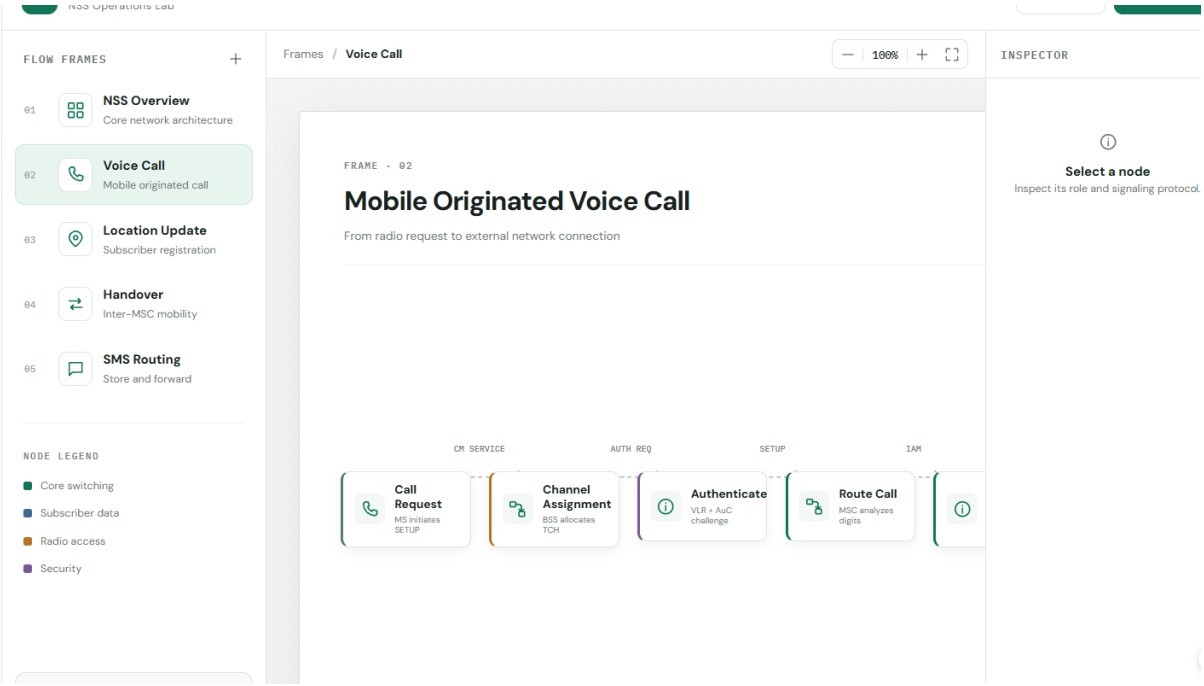
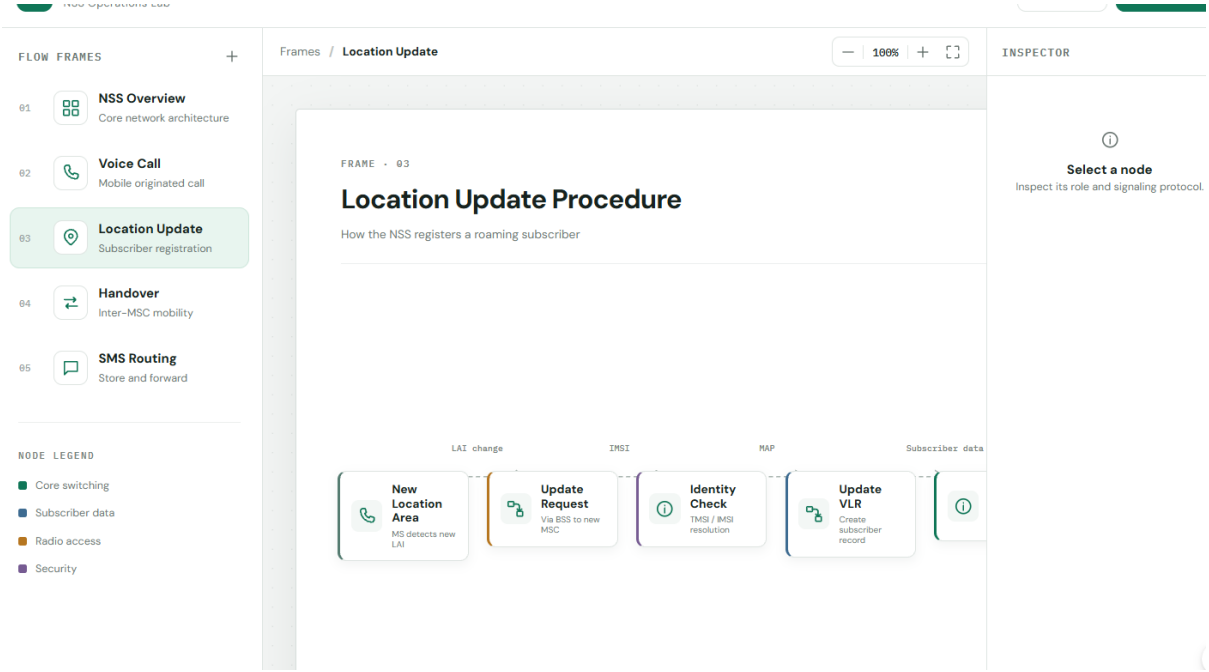
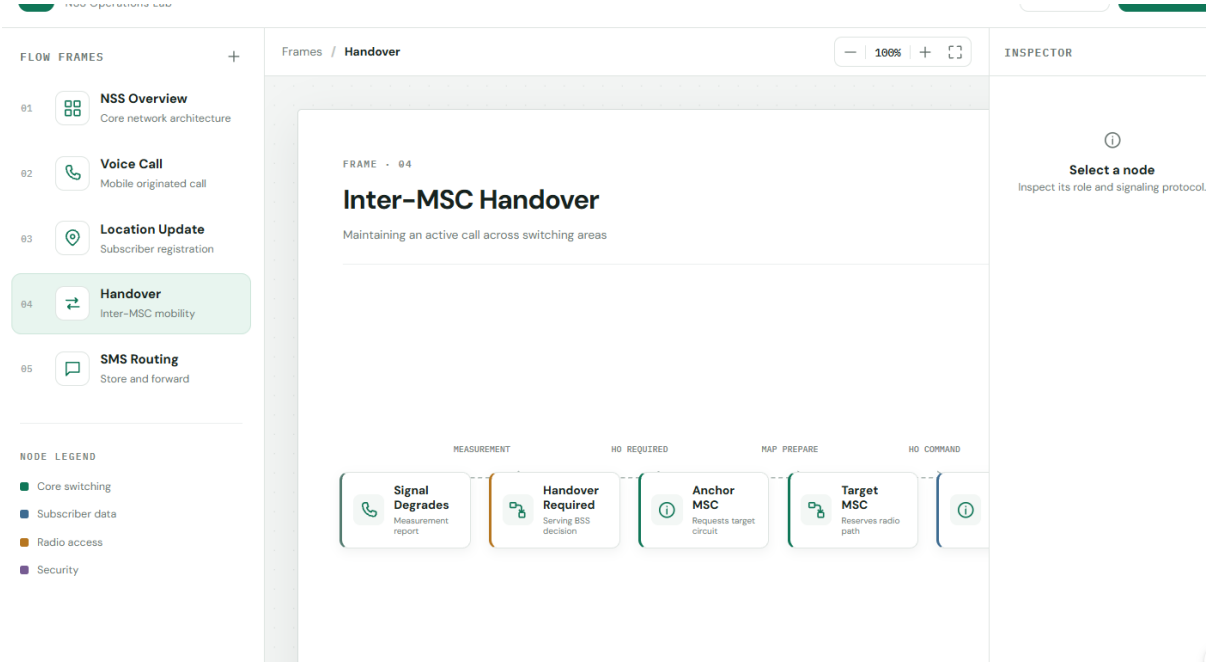
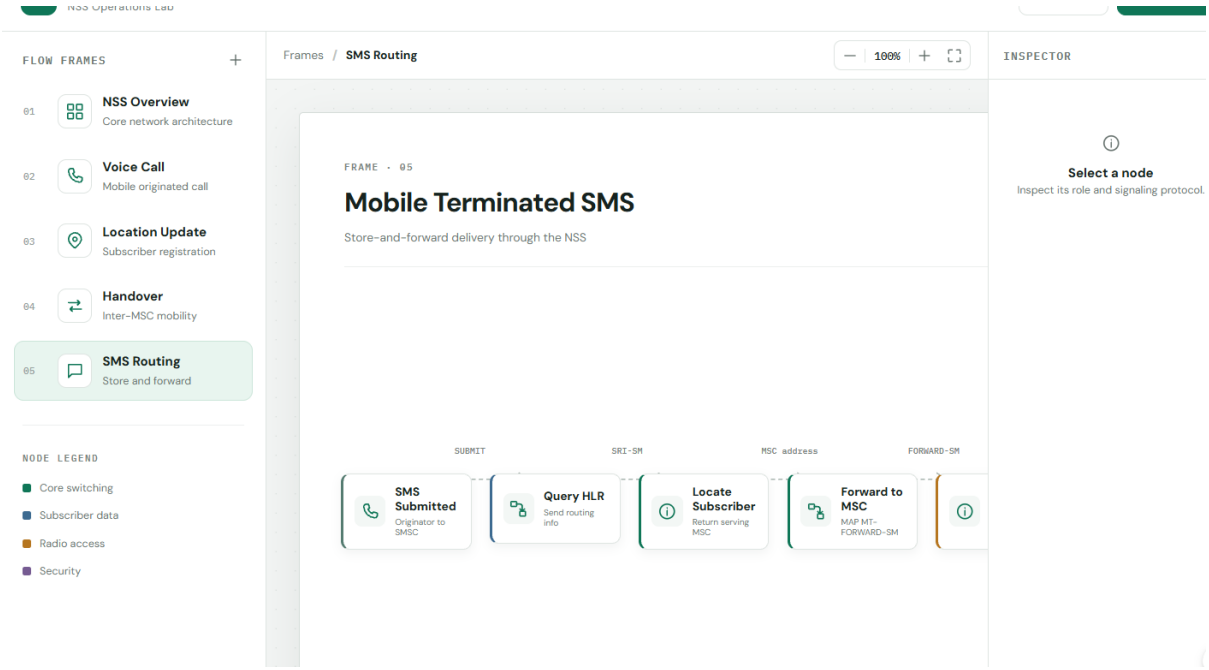
Experiment 3






FLOW FRAMES
01 NSS Overview
Core network architecture
02 Voice Call
Mobile originated call
03 Location Update
Subscriber registration
04 Handover
Inter-MSC mobility
05 SMS Routing
Store and forward
NODE LEGEND
Core switching
Subscriber data
Radio access
Security
Frames / Location Update
100%
INSPECTOR
FRAME · 03
Location Update Procedure
How the NSS registers a roaming subscriber
LAI change
IMSI
MAP
Subscriber data
New Location Area
MS detects new LAI
Update Request
Via BSS to new MSC
Identity Check
TMSI / IMSI resolution
Update VLR
Create subscriber record
Select a node
Inspect its role and signaling protocol.


FLOW FRAMES
01 NSS Overview
Core network architecture
02 Voice Call
Mobile originated call
03 Location Update
Subscriber registration
04 Handover
Inter-MSC mobility
05 SMS Routing
Store and forward
NODE LEGEND
Core switching
Subscriber data
Radio access
Security
Frames / Handover
100%
INSPECTOR
FRAME · 04
Inter-MSC Handover
Maintaining an active call across switching areas
MEASUREMENT
HO REQUIRED
MAP PREPARE
HO COMMAND
Signal Degrades
Measurement report
Handover Required
Serving BSS decision
Anchor MSC
Requests target circuit
Target MSC
Reserves radio path
Select a node
Inspect its role and signaling protocol.


FLOW FRAMES
01
NSS Overview
Core network architecture
02
Voice Call
Mobile originated call
03
Location Update
Subscriber registration
04
Handover
Inter-MSC mobility
05
SMS Routing
Store and forward
NODE LEGEND
Core switching
Subscriber data
Radio access
Security
Frames / SMS Routing
100%
INSPECTOR
FRAME · 05
Mobile Terminated SMS
Store-and-forward delivery through the NSS
SUBMIT
SRI-SM
MSC address
FORWARD-SM
SMS Submitted
Originator to SMSC
Query HLR
Send routing info
Locate Subscriber
Return serving MSC
Forward to MSC
MAP MT-FORWARD-SM
Select a node
Inspect its role and signaling protocol.